

Simulated Annealing Algorithm

Problem: Maximize $f(x) = (\cos(50x) + \sin(20x))^2$, $x \in (0,1)$

Algorithm: Simulated Annealing

Input: N (iterations), r (step size), f (objective function)

Output: x^* (best solution found)

Step 1: Initialize

$$x_1 \leftarrow \text{Uniform}(0,1)$$

Step 2: For $k = 2$ to N :

(a) Generate candidate: $a \leftarrow \text{Uniform}(x_{k-1} - r, x_{k-1} + r)$

(b) Temperature schedule: $T_k \leftarrow 1/\ln(k)$

(c) Acceptance ratio: $\alpha \leftarrow e^{(f(a) - f(x_{k-1}))/T_k}$

(d) Accept criterion:

if $u < \min(1, \alpha)$ where $u \sim \text{Uniform}(0,1)$:

$$x_k \leftarrow a$$

else:

$$x_k \leftarrow x_{k-1}$$

Step 3: Return $x^* = \arg \max\{f(x_i)\}$ for $i = 1$ to N

Key Components:

1. Objective Function (Transformed): $h(x,T) = e^{f(x)/T}$

- High T: Flatter landscape $\rightarrow$ Exploration
- Low T: Sharp peaks $\rightarrow$ Exploitation

2. Cooling Schedule: $T_k = 1/\ln(k)$

- Sub-geometric cooling
- Ensures: Early iterations explore, later iterations refine

3. Metropolis Acceptance:

- If $\alpha > 1$: Accept always (uphill move)
- If $\alpha \leq 1$: Accept with probability α (downhill move)
- Temperature controls tolerance for worse solutions

4. Random Walk Neighborhood: $a \leftarrow \text{Uniform}(x_{k-1} - r, x_{k-1} + r)$

- $r = 0.3$ controls step size
- Ensures local connectivity

Results on $f(x) = (\cos(50x) + \sin(20x))^2$:

Temperature Effect (From Plot):

- $T = 1.00$: Flattened peaks, broad exploration
- $T = 0.83$: Moderate peak sharpening
- $T = 0.75$: Further sharpening
- $T = 0.71$: Sharp peaks, localized search

Algorithm Execution (N = 500):

- Best solution found: $x^* \approx 0.3792$
- Function value: $f(x^*) \approx 0.96$
- Trajectory: Points cluster around major peaks
- Algorithm successfully escaped local maxima

Convergence Properties:

- Accepts better solutions always: $\alpha > 1$ ■ uphill bias
- Accepts worse solutions probabilistically: allows escape from local optima
- Temperature-dependent acceptance: $P(\text{accept}|\text{downhill}) = e^{\Delta f/T}$
- Sub-geometric schedule: Ensures convergence to global optimum as $N \rightarrow \infty$
- Trade-off: Early iterations explore broadly; later iterations refine locally